Nakyung (Chloe) Oh

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EDUCATION

M.Sc. Psychology: Data Science in Human Behavior Expected: Dec 2026
University of Wisconsin-Madison, Madison, WI

B.Sc. Psychology, Minor in Statistics Dec 2024
University of Cincinnati, Cincinnati, OH | GPA: 3.809

TECHNICAL SKILLS

Programming and data: Python, R, SQL, SAS, Excel, Google Sheets
Methods: K-nearest neighbors, neural networks, classification, regression, model evaluation, behavioral data analysis
Research and analytic tools: API, prompt design, R Studio, JASP, GraphPad Prism, ImageJ, BORIS, Jupyter Notebook, Jupyter Lab
Language: Korean (native), English (fluent)

DATA PROJECTS

Reading Ability and Mathematics Performance Analysis Sep 2026 - Present
Data Science Research Project | Carnegie Learning

- Analyze relationships between predicted reading ability and mathematics performance using mixed-effects logistic regression, accounting for student and lesson level differences
- Identify that a 0.10 increase in predicted reading ability was associated with 30% higher odds of answering math questions correctly and approximately 15% higher odds of lesson completion
- Investigate variations in reading ability effects across mathematics lessons and content types to identify potential learning barriers

Historical Document Transcription Sep 2026 - Present
Machine Learning Project | UW Madison Machine Learning Marathon

- Develop a Python pipeline using CHURRO-3B and Qwen vision-language models to transcribe historical handwritten documents
- Evaluate preprocessing and prompting strategies to improve handwriting recognition, using character error rate (CER) to compare predictions with reference transcriptions
- Report a preliminary CER of 7.65% across 2,751 handwritten German text lines, with ongoing model refinement to improve transcription accuracy

Mr. Macky's Data Analysis with Excel Sep 2026 - Present
Analytical Project | Instructor: Dr. Barry Gerhart

- Analyze employee turnover, engagement, and sales performance across 126 stores using Excel, pivot tables, correlation analysis, and regression
- Detect strong annual turnover consistency ($r = 0.70$) and an association between engagement and turnover ($r = 0.79$)
- Demonstrate that a 10 percentage point increase in turnover was associated with approximately $150,000 lower annual store sales, controlling for prior year sales

LLM Assisted Analysis of Goodreads Review Mar 2026 - Present
Research Project | Mentor: Dr. Nick Buttrick

- Investigate reading experiences and psychological complexity using a Goodreads dataset containing over 100 million reviews
- Utilize LLM to classify Goodreads reviews across seven psychological and literary constructs, refining prompts to improve classification accuracy and consistency
- Evaluate LLM classifications against 96 human-annotated reviews, achieving weighted F1 scores of 0.752-0.939 across seven constructs

fMRI Emotion Regulation Classification Mar 2026 - May 2026

Machine Learning Project | Mentor: Dr. João Guassi Moreira

- Developed a K-nearest neighbors (KNN) classifier to distinguish emotion regulation conditions from fMRI data across 200 brain parcels
- Applied principal component analysis (PCA) for dimensionality reduction and evaluated classification performance across brain regions
- Achieved 75% classification accuracy, the highest among project team members

Neural Network Model Reimplementation Jan 2026 - May 2026

Computational modeling project | Instructor: Dr. Tim Rogers

- Reimplemented the Rumelhart and McClelland (1986) past-tense learning model in Python, reproducing overregularization and U-shaped learning patterns
- Evaluated performance across 502 verbs and frequency groups, achieving 96.2% training accuracy and 88.4% accuracy on low-frequency test verbs

RELEVANT EXPERIENCE

Data Collector Dec 2025

Department of Life Science Communication, University of Wisconsin-Madison

- Researched and verified 500 mailing addresses for AI researchers using institutional websites, Google Scholar, and other public sources
- Organized and validated researcher information in MS Excel to ensure accurate and consistent records

Research Assistant May 2024 - Dec 2024

Ulrich-Lai Laboratory, UC Metabolic Diseases Institute

- Coded rodent social interaction behaviors using BORIS and performed statistical analyses with GraphPad Prism
- Quantified perineuronal net (PNN) and parvalbumin fluorescence intensity in prefrontal cortex images using ImageJ

Research Assistant Sep 2023 - Aug 2024

Children Cognitive Research Lab, University of Cincinnati

- Collected, scored, and organized children's cognitive and mathematics assessment data from Corryville Elementary School
- Collaborated with researchers to validate assessment scores and prepare structured datasets using Google Sheets